

NOETED

Public Comment — Request for Information

Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Docket No. HHS-ONC-2026-0001 | Submitted February 2026

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About NOETED:

AI-powered healthcare compliance intelligence platform. Member, NVIDIA Inception Program. Building centralized regulatory intelligence infrastructure for U.S. healthcare organizations.

Preamble

NOETED is an AI-powered healthcare compliance intelligence platform founded in Redding, California. We are building the centralized regulatory intelligence infrastructure that U.S. healthcare organizations currently lack: a single system that continuously monitors, aggregates, and connects compliance requirements from federal agencies (CMS, OIG, FDA, SAMHSA, ONC), state health departments, and regional regulatory bodies — delivering real-time alerts and AI-assisted gap analysis to healthcare payers, providers, and health plans.

NOETED is a member of the NVIDIA Inception Program. Our perspective on this RFI is informed by direct experience building AI systems designed to operate within the U.S. healthcare regulatory environment — and by the specific infrastructure barriers we have encountered in attempting to access the federal and state regulatory data that healthcare compliance depends on.

We respond to each of HHS's specific questions below. A consistent theme runs through our responses: the regulatory data infrastructure problem. The APIs, databases, and publication systems through which federal and state agencies publish compliance requirements are frequently antiquated, unreliable, or nonexistent. This infrastructure failure is not peripheral — it is the foundational barrier that prevents AI compliance tools from functioning reliably, that forces healthcare organizations into manual monitoring processes, and that ultimately harms patients when compliance failures result in disrupted care. We believe solving this infrastructure problem is the highest-leverage action HHS can take to accelerate responsible AI adoption in clinical care.

NOETED Response

From NOETED's direct experience building AI for healthcare compliance, we identify four primary barriers:

1. Regulatory fragmentation and inaccessibility.

The most underappreciated barrier to healthcare AI adoption is not the AI itself — it is the compliance environment the AI must operate within. Healthcare organizations attempting to deploy AI tools face requirements originating from CMS conditions of participation, OIG compliance program guidance, FDA software-as-a-medical-device rules, state insurance commissioner mandates, HIPAA privacy and security rules, 21st Century Cures Act information blocking provisions, and ONC's HTI-1 AI transparency requirements. Each regulatory body publishes changes through different channels, on different schedules, in different formats. There is no centralized, machine-readable, authoritative source of healthcare regulatory requirements. This creates a compliance environment that is functionally inaccessible to organizations without large legal teams — and that directly chills AI innovation because organizations cannot assess whether their AI tools will be compliant with requirements they cannot reliably track.

2. Reimbursement uncertainty.

Private sector AI developers cannot sustain investment in clinical AI tools without a clear path to reimbursement. The absence of consistent Medicare and Medicaid payment pathways for AI-assisted clinical services creates a market disincentive that is particularly acute for companies serving smaller providers and rural health systems, where the cost of regulatory compliance and the uncertainty of reimbursement combine to make the investment calculus prohibitive.

3. Liability ambiguity for AI-assisted clinical decisions.

The existing legal framework does not clearly allocate liability when an AI tool contributes to a clinical decision that results in patient harm. This ambiguity discourages both AI developers (who face unpredictable risk) and healthcare organizations (who cannot obtain clear indemnification). Until this framework is clarified, risk-averse healthcare systems will default to limiting AI deployment to low-stakes administrative functions rather than the high-value clinical applications that could most improve outcomes.

4. Broken regulatory data infrastructure.

AI systems require high-quality, structured, standardized data. Despite progress under the 21st Century Cures Act and ONC interoperability rules, significant barriers remain in the regulatory data domain specifically. NOETED has encountered this infrastructure failure directly: the APIs, databases, and publication systems through which federal and state agencies publish compliance requirements are frequently antiquated, unreliable, or nonexistent. Where APIs do exist, access requirements are burdensome, inconsistently documented, and subject to approval processes with no defined timeline or accountability. Datasets are poorly organized, infrequently updated, and structurally inconsistent across agencies and jurisdictions. Regulatory data that is legally public and operationally critical to compliance monitoring is in practice inaccessible due to broken infrastructure and deprecated endpoints that no agency has been directed to maintain.

This is not a peripheral inconvenience. It is a foundational infrastructure failure that prevents AI compliance tools from delivering reliable intelligence, forces healthcare organizations to rely on manual monitoring processes, and directly harms patients when the resulting compliance failures disrupt access to care. Smaller organizations — FQHCs, rural hospitals, community health centers — are disproportionately affected because they lack the technical capacity to work around broken infrastructure that larger systems can navigate through dedicated IT staff.

Q2

What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care? What HHS regulations, policies, or programs could be revisited? Please provide specific changes and applicable CFR citations.

NOETED Response

NOETED recommends the following specific changes, organized by regulatory domain. Our recommendations are informed by direct experience building AI-powered regulatory intelligence infrastructure and by the specific barriers NOETED has encountered accessing the federal and state regulatory data that healthcare compliance depends on.

A. Mandate modernization of federal regulatory data publication infrastructure

This is NOETED's highest-priority recommendation because it is the foundational barrier that makes every other recommendation harder to implement.

NOETED has encountered firsthand the infrastructure failure at the core of healthcare regulatory data: the APIs, databases, and publication systems through which federal and state agencies publish compliance requirements are frequently antiquated, unreliable, or nonexistent. Where APIs do exist, access requirements are burdensome, inconsistently documented, and subject to approval processes with no defined timeline or accountability. Datasets are poorly organized, infrequently updated, and structurally inconsistent across agencies and jurisdictions. Regulatory data that is legally public and operationally critical to compliance monitoring is in practice inaccessible — not because of policy decisions, but because of broken infrastructure and deprecated endpoints that no agency has been directed to maintain.

This is not a peripheral inconvenience. It is a foundational infrastructure failure that prevents AI compliance tools from delivering reliable intelligence and forces healthcare organizations to continue relying on manual, error-prone monitoring processes — exactly the processes that produce the compliance failures that harm patients and cost the system billions annually.

HHS should issue a directive requiring all agencies publishing healthcare regulatory data — including CMS, OIG, FDA, SAMHSA, and ONC — to meet minimum technical publication standards within a defined implementation timeline. Those standards should include active, documented, versioned APIs for all regulatory data; machine-readable publication formats with consistent schema; defined update frequency requirements tied to regulatory change events; and a designated point of contact within each agency responsible for regulatory data infrastructure.

Relevant authority: 45 CFR Part 170 (ONC Health IT Certification) — ONC should extend certification criteria to include standardized regulatory data publication APIs, consistent with existing FHIR API requirements under 45 CFR § 170.315(g)(10). HHS should additionally act under the Evidence Act (Pub. L. 115-435) and OMB Circular A-130 to require open, machine-readable regulatory data publication across all HHS divisions as a condition of federal information management compliance.

B. Establish a standardized regulatory data publication framework (ONC / HHS-wide)

Modernizing the infrastructure addresses access. Standardizing the format addresses usability. Both are required.

Currently, CMS, OIG, and state agencies publish regulatory changes through inconsistent channels — Federal Register notices, agency website postings, email listservs, and PDF documents — with no common data model, no structured metadata, no version control, and no programmatic access. A compliance technology platform attempting to monitor regulatory changes across even a subset of federal sources must manage dozens of incompatible formats, inconsistent publication schedules, and no reliable mechanism for detecting when a source has been updated.

HHS should require all agencies publishing healthcare compliance requirements to adopt a machine-readable, standardized format for regulatory notifications — analogous to what HL7 FHIR has accomplished for clinical data. This standard should include mandatory metadata fields: effective date, affected organization types, applicable CFR citations, superseded guidance references, and a unique identifier for each regulatory change event.

Relevant CFR: 45 CFR Part 170 (ONC Health IT Certification) — ONC should extend certification criteria to include standardized regulatory data publication APIs. Additionally, ONC should commission a regulatory data FHIR Implementation Guide, establishing a FHIR-compatible specification for publishing healthcare regulatory requirements through existing health IT infrastructure.

C. Create consistent Medicare reimbursement pathways for AI-assisted clinical services

CMS should establish durable, technology-neutral reimbursement codes for AI-assisted clinical decision support, rather than relying on coverage decisions that vary by contractor and are subject to frequent revision. The current approach creates the reimbursement uncertainty that suppresses private sector AI investment — particularly for companies serving smaller providers and rural health systems, where the cost of regulatory compliance and the uncertainty of reimbursement combine to make the investment calculus prohibitive.

Reimbursement pathways must be durable. Coverage decisions that can be reversed by a single contractor determination create the kind of investment uncertainty that drives AI developers toward larger, better-resourced markets and away from the rural and safety-net settings where AI could have the greatest equity impact.

Relevant CFR: 42 CFR Part 414 (Medicare Part B Payment) — CMS should issue rulemaking establishing a consistent, technology-neutral payment framework for AI-assisted clinical services with clear criteria for what constitutes a reimbursable AI-assisted service under § 414.1 et seq. Contractor discretion in coverage determination for AI-assisted services should be constrained by national coverage policy.

D. Clarify OIG safe harbor treatment for AI tool investments

The Anti-Kickback Statute (42 U.S.C. § 1320a-7b(b)) and associated OIG safe harbors create ambiguity around whether health system investments in AI tools — particularly when provided to referring providers — constitute prohibited remuneration. This ambiguity has a chilling effect on the formation of health system and AI developer partnerships that are necessary for AI tools to reach smaller providers at scale.

Relevant CFR: 42 CFR § 1001.952 (OIG Safe Harbors) — OIG should add an explicit AI technology safe harbor modeled on the existing Electronic Health Records safe harbor at § 1001.952(y), covering the provision of AI clinical decision support tools to referring providers under defined conditions including fair market value cost-sharing and non-exclusivity requirements.

E. Expand LEAP in Health IT to include regulatory intelligence infrastructure

ONC's Leading Edge Acceleration Projects program has historically funded innovative health IT solutions at the frontier of interoperability. NOETED respectfully and specifically requests that ONC include regulatory intelligence infrastructure — systems that help healthcare organizations continuously monitor and respond to compliance requirements across federal, state, and local regulatory sources — as an eligible and priority area of interest in future LEAP funding cycles.

This is not a peripheral use case. Regulatory intelligence infrastructure is a prerequisite for trustworthy AI deployment in healthcare. An AI clinical decision support tool that operates in a compliance environment the deploying organization cannot monitor is not a safe tool. Funding the infrastructure layer that makes compliance monitoring possible is as foundational as funding the clinical AI tools that depend on it.

Relevant program: ONC LEAP (NAP-AX-22-001) — no CFR citation required. This is a programmatic recommendation for ONC's discretionary funding priorities under the existing LEAP cooperative agreement framework.

F. Revise HIPAA guidance to clarify AI training data use

The current HIPAA Privacy Rule creates uncertainty around the permissible use of protected health information for AI model training, even when data is de-identified under Safe Harbor or Expert Determination methods. This uncertainty is compounded by the regulatory data access barriers NOETED has encountered directly: organizations that want to build AI systems using de-identified regulatory and compliance data face not only HIPAA ambiguity but also the practical inaccessibility of the underlying datasets described in Recommendation A.

HHS should issue updated guidance clarifying the conditions under which de-identified healthcare data may be used for AI training without individual authorization, and should explicitly address synthetic data generation as an alternative to de-identified PHI for AI training — an increasingly viable approach that current HIPAA guidance does not clearly address.

Relevant CFR: 45 CFR § 164.514 (De-identification of PHI) — HHS should issue updated guidance under § 164.514 specifically addressing AI training data use, including standards for de-identification sufficiency in machine learning contexts and safe harbor treatment for synthetic data generation methods that do not use PHI as direct training input.

NOETED Response**Liability allocation in AI-assisted clinical decisions.**

When an AI tool contributes to a clinical decision that results in patient harm, existing tort law does not clearly allocate liability among the AI developer, the healthcare organization that deployed the tool, the clinician who relied on it, and the data providers whose data trained it. This multi-party liability problem is novel and existing medical malpractice frameworks were not designed to address it. HHS should work with the Department of Justice and state attorneys general to develop a model liability framework that clarifies roles and responsibilities across the AI deployment chain, provides safe harbors for organizations that follow defined best practices, and creates a consistent national standard rather than allowing 50 divergent state-law approaches to develop simultaneously.

Regulatory compliance liability for AI-generated recommendations.

NOETED's direct experience reveals a specific and underappreciated liability issue: when AI systems generate compliance recommendations that prove incorrect, the question of who bears liability for the resulting regulatory violation is unresolved. A healthcare organization that relies on an AI compliance tool and subsequently fails a CMS audit cannot currently point to the AI tool's error as a defense — they remain liable for the underlying compliance failure regardless of the tool's role. HHS should issue guidance clarifying how AI-assisted compliance errors are treated in enforcement contexts, and whether demonstrated reliance on a certified AI compliance tool affects penalty calculation under 42 CFR Part 1003 (OIG Civil Monetary Penalties).

Source data liability — a novel issue unique to regulatory AI.

A third liability issue that existing legal frameworks do not address is source data liability for AI compliance tools. NOETED has documented firsthand that federal and state agencies publish regulatory data through antiquated, unreliable infrastructure — broken APIs, deprecated endpoints, inconsistently updated datasets. When an AI compliance tool fails because the underlying regulatory data published by a federal agency was incorrect, outdated, or inaccessible, the liability chain becomes deeply unclear. The healthcare organization relied on the AI tool; the AI tool relied on the agency-published data; the agency published data through infrastructure it has not maintained. No existing legal framework allocates liability across this chain.

This is not a hypothetical. An AI compliance tool trained on CMS data that has not been updated to reflect a regulatory change will generate recommendations that are technically wrong — and a healthcare organization following those recommendations may fail a CMS audit for a violation it had no practical means of anticipating. HHS should address this gap directly: agencies that publish regulatory data through broken or unreliable infrastructure bear some measure of responsibility for downstream compliance failures attributable to that infrastructure failure. HHS should issue guidance clarifying the government's obligations regarding the reliability and currency of publicly published regulatory data, and the extent to which reliance on official government data sources affects enforcement discretion in CMS and OIG proceedings.

Privacy implications of AI systems that process unstructured clinical data.

AI systems operating in clinical environments frequently process unstructured data — clinical notes, voice recordings, imaging reports — that contains PHI in forms not anticipated by existing HIPAA

guidance. The intersection of large language model processing with HIPAA's minimum necessary standard (45 CFR § 164.502(b)) creates compliance questions that current guidance does not resolve. HHS should issue a specific guidance document addressing AI processing of unstructured clinical data, including requirements for data minimization, retention, and audit logging in AI systems handling PHI.

Recommended HHS role.

HHS should convene a multi-stakeholder working group — including AI developers, healthcare organizations, legal scholars, and patient advocates — to develop model governance frameworks for AI in clinical care. These frameworks should address liability allocation across the full AI deployment chain including source data publishers, indemnification standards, privacy safeguards for AI processing, and security requirements for AI systems with access to PHI. ONC's existing Health IT certification program could serve as the vehicle for implementing minimum governance standards for AI tools used in certified EHR contexts.

Q4

For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and human-centered evaluation methods for clinical care? Should HHS support these? If so, which mechanisms?

NOETED Response

Pre-deployment evaluation.

The most promising pre-deployment evaluation frameworks for clinical AI share three characteristics: they test performance across demographically diverse subpopulations rather than aggregate populations only; they include adversarial robustness testing against distribution shift (i.e., how the model performs when real-world data differs from training data); and they incorporate structured clinical expert review of model outputs before deployment, not just statistical performance metrics. NOETED recommends that HHS support the development of standardized pre-deployment evaluation protocols specific to healthcare AI use cases, including compliance decision support, clinical documentation, and diagnostic assistance.

A critical and currently unaddressed dimension of pre-deployment evaluation for compliance AI is source data quality assessment. Evaluation frameworks assume that training and validation data accurately represents the ground truth. For clinical AI operating in well-defined domains this assumption is reasonable. For AI compliance tools, it is not — because the ground truth is regulatory data published by federal and state agencies through infrastructure that NOETED has documented to be antiquated, inconsistent, and unreliable. HHS should require that pre-deployment evaluation of AI compliance tools include explicit assessment of the quality, currency, and completeness of the regulatory data sources the tool depends on. An AI compliance tool cannot be evaluated independently of the data infrastructure it relies on.

Post-deployment monitoring.

Post-deployment monitoring is where current practice is weakest. Most healthcare AI deployments lack systematic mechanisms for detecting model drift, bias emergence, or performance degradation over time. HHS should require post-deployment monitoring plans as a condition of any federal funding for AI deployment, and should support the development of automated monitoring tools that can detect performance changes without requiring manual review of every AI output. For AI compliance tools specifically, post-deployment monitoring must include tracking of regulatory source data currency — a tool that was accurate at deployment may become inaccurate when the underlying regulatory data changes and the tool is not updated.

Human-centered evaluation.

Evaluation methods that measure only technical performance miss the question of whether AI tools actually improve clinical workflows and outcomes. HHS should prioritize funding for human-centered evaluation research — studies that measure how clinicians actually interact with AI tools, where AI recommendations are accepted or overridden and why, and what the downstream patient outcome effects are. This research is currently sparse and is essential for understanding whether AI tools that perform well in technical evaluations translate to real-world clinical benefit.

Recommended HHS mechanisms.

NOETED recommends cooperative agreements as the primary mechanism, given their flexibility and ability to support multi-site, multi-stakeholder evaluation projects. Prize competitions are appropriate for well-defined technical challenges such as best-performing model on a specific benchmark dataset.

Grants are appropriate for foundational research on evaluation methodology. HHS should also consider creating a publicly accessible Healthcare AI Evaluation Repository — a centralized database of evaluation results for deployed clinical AI tools — that would allow healthcare organizations to make informed procurement decisions.

NOETED Response**Extend ONC Health IT Certification to AI compliance tools.**

ONC's existing Health IT Certification Program (45 CFR Part 170) provides a proven framework for establishing minimum standards for health IT tools. HHS should extend this framework to include certification criteria for AI tools used in clinical care — particularly AI compliance and decision support tools that are not classified as medical devices but that directly influence clinical operations. Certification should require demonstrated performance across diverse patient populations, documented bias testing, interoperability with certified EHRs, and ongoing post-deployment monitoring.

Support industry-driven credentialing with federal recognition.

Industry consortia — including HIMSS, AHIMA, and emerging AI-specific organizations — are developing voluntary credentialing frameworks for healthcare AI. HHS should establish a formal recognition process for these industry-driven frameworks, similar to how NCQA accreditation is recognized in Medicare and Medicaid contracting. Federal recognition creates market incentives for voluntary credentialing adoption without requiring prescriptive federal rulemaking.

Create a safe harbor for certified AI tools in enforcement contexts.

HHS should establish that healthcare organizations using AI tools that have achieved ONC certification or HHS-recognized industry credentialing receive defined safe harbor treatment in CMS and OIG enforcement proceedings arising from AI-assisted compliance errors. This directly addresses the liability concern identified in Q3 and creates a powerful market incentive for both AI developers to seek certification and healthcare organizations to use certified tools.

Regulatory intelligence as a certification prerequisite — contingent on infrastructure modernization.

NOETED specifically recommends that ONC require certified AI tools to demonstrate integration with a standardized regulatory notification system as a certification prerequisite. An AI clinical decision support tool that cannot monitor its own compliance environment — one that cannot automatically update its logic when the regulations governing it change — is not a trustworthy tool. Regulatory intelligence integration should be a certification criterion, not an optional feature.

This recommendation is directly dependent on Recommendation A in Q2: the certification prerequisite is only implementable if the federal regulatory data infrastructure exists and is reliable. ONC cannot require AI tools to integrate with a regulatory notification system that does not yet exist in machine-readable form. HHS must therefore treat infrastructure modernization and certification criteria development as a sequenced initiative — the infrastructure must be built before the certification requirement can be enforced. The two recommendations are linked and must be pursued together.

Q6

Where have AI tools in clinical care met or exceeded performance and cost expectations, and where have they fallen short? What novel AI tools would have the greatest potential to improve health care outcomes, quality insights, and cost reduction?

NOETED Response

Where AI has met or exceeded expectations.

The strongest evidence of AI performance meeting clinical expectations is in ambient clinical documentation — AI-assisted transcription and documentation tools that reduce physician administrative burden. These tools have achieved high adoption rates in some health systems because they address a clear, painful, well-defined problem without requiring AI to make clinical judgments. Similarly, AI-assisted imaging analysis (radiology, pathology) has shown strong performance in narrow, well-defined classification tasks where ground truth is clear and training data is abundant.

Where AI has fallen short.

AI tools have consistently underperformed in contexts requiring generalization across diverse patient populations, particularly when training data underrepresents minority and rural populations. AI tools have also fallen short in dynamic regulatory environments — and NOETED's direct experience building in this space reveals two distinct failure modes that are frequently conflated but require separate solutions.

The first failure mode is update lag: tools trained on historical compliance standards fail when regulations change and the tools are not updated to reflect new requirements. This is a development and maintenance problem that better engineering practices can address.

The second failure mode is infrastructure failure: tools fail not because their developers failed to update them, but because the underlying regulatory data infrastructure they depend on — the government-published APIs, databases, and notification systems — is itself broken, inconsistent, or inaccessible. A compliance AI tool cannot be current if the data source it monitors is not current. This is not a problem AI developers can solve unilaterally. It requires the federal regulatory data infrastructure modernization described in Q2 Recommendation A. Until that infrastructure exists, AI compliance tools will continue to fail in the field not due to any deficiency in the AI itself, but due to deficiencies in the government data systems feeding it.

Novel tools with the greatest potential.

NOETED identifies the following as highest-potential areas for novel AI tools in healthcare:

- Regulatory intelligence and compliance monitoring — AI that continuously tracks regulatory changes and maps their implications across the compliance obligations of a specific healthcare organization. This is infrastructure-level AI that enables every other clinical AI tool to operate safely. Its effectiveness depends directly on the quality of the regulatory data infrastructure described throughout this comment.
- Predictive compliance risk scoring — AI that identifies healthcare organizations at elevated risk of compliance failure before enforcement actions occur, enabling proactive intervention.
- AI-assisted prior authorization — automating the clinical criteria review process that currently consumes enormous administrative resources and creates delays in patient care.

- Population health compliance monitoring — AI that identifies compliance gaps in care delivery for specific patient populations, particularly for value-based care quality measures.
- Rural and FQHC-specific clinical AI — tools designed for the resource constraints and patient populations of rural and community health settings, where AI deployment has lagged due to infrastructure and cost barriers.

NOETED Response

Most influential decision makers for AI adoption.

Based on NOETED's engagement with healthcare organizations, the decision makers with the greatest influence on AI adoption decisions are, in order of impact:

1. Chief Compliance Officers (CCOs) and Chief Legal Officers (CLOs) — who must approve any AI tool that touches compliance-sensitive workflows. CCOs are the gatekeepers for healthcare AI because almost all clinical AI tools have compliance implications. Their concerns center on regulatory risk, liability, and auditability.
2. Chief Information Officers (CIOs) and Chief Medical Information Officers (CMIOs) — who evaluate technical integration, data security, and EHR compatibility. Tools that cannot integrate with existing EHR infrastructure typically fail at this stage regardless of clinical merit.
3. Chief Financial Officers (CFOs) — who evaluate reimbursement viability and ROI. The absence of consistent reimbursement pathways identified in Q2 means CFOs frequently cannot build a viable business case for AI tool investment.
4. Clinical champions — individual physicians or department heads who advocate for specific AI tools based on clinical experience. Without a clinical champion, AI tools typically cannot advance past pilot stage.

Primary administrative hurdles.

The administrative hurdles NOETED has observed most frequently are:

- Compliance review paralysis — the compliance review process for new AI tools has no standardized framework, leading to ad hoc, organization-specific reviews that can take 6–18 months and frequently result in tools being rejected due to unresolved regulatory ambiguity rather than actual compliance failure. This problem is compounded when the AI tool in question is a compliance tool itself: CCOs reviewing an AI compliance tool cannot meaningfully assess the tool's reliability when the underlying regulatory data the tool depends on is itself unreliable or inaccessible — a direct consequence of the infrastructure failure described in Q1 and Q2.
- EHR vendor lock-in — AI tools that are not part of the incumbent EHR vendor's ecosystem face significant integration barriers, including API restrictions, data access limitations, and EHR vendor contract provisions that restrict third-party integrations.
- Procurement complexity — healthcare organizations subject to federal contracting requirements face procurement processes designed for physical goods, not software-as-a-service AI tools. These processes create delays and documentation burdens that disproportionately burden smaller AI developers.
- Staff training and change management — even technically successful AI deployments frequently fail in practice due to inadequate training, workflow integration, and change management support. AI developers who are not resourced to provide this support see adoption collapse after initial deployment.

Q8

Where would enhanced interoperability widen market opportunities, fuel research, and accelerate AI development for clinical care? Consider specific data types, data standards, and benchmarking tools.

NOETED Response

The regulatory data gap — NOETED's firsthand account.

The healthcare interoperability conversation has focused almost exclusively on clinical data (patient records, lab results, imaging) and administrative data (claims, prior authorizations). NOETED has identified a third critical data domain that is almost entirely absent from interoperability frameworks and that we have experienced as the primary technical obstacle to building reliable healthcare AI: regulatory data — the structured, machine-readable publication of compliance requirements, regulatory changes, and enforcement guidance.

In building NOETED's compliance intelligence platform, we have attempted to programmatically access regulatory data from CMS, OIG, state health departments, and other agencies. What we found is a landscape of broken APIs with no documented maintenance schedules, datasets that had not been updated to reflect regulatory changes that occurred months prior, access requirement processes with no defined timelines, inconsistent data schemas across agencies publishing information about the same regulatory domain, and in several cases, APIs that were listed in official documentation but returned errors or no data when called. This is the regulatory data infrastructure that every healthcare AI compliance tool must navigate. It is not navigable at scale without manual intervention — which defeats the purpose of AI automation entirely.

Enhanced interoperability in the regulatory data domain would have a greater near-term impact on healthcare AI development than equivalent investment in clinical data interoperability, because the clinical data infrastructure — while imperfect — is dramatically more mature. The regulatory data infrastructure barely exists.

Specific data types where enhanced interoperability would have the greatest impact.

- Regulatory change notifications — standardized, machine-readable publication of all CMS, OIG, FDA, and ONC regulatory changes, with metadata indicating affected organization types, effective dates, and applicable CFR citations.
- Compliance program guidance — OIG's compliance program guidance documents, currently published as static PDFs, should be converted to structured, queryable formats with version history and change tracking.
- Enforcement action data — CMS and OIG enforcement actions, currently scattered across multiple databases, should be consolidated into a single, structured, queryable dataset that AI systems can use for compliance risk modeling.
- Quality measure specifications — CMS quality measure specifications, which change annually, should be published in machine-readable formats that AI clinical decision support tools can ingest without manual programming updates.

Recommended standards and benchmarking tools.

HHS should direct ONC to develop a regulatory data FHIR Implementation Guide — a standardized specification for publishing healthcare regulatory requirements in FHIR-compatible formats. This would allow existing health IT infrastructure to serve as the distribution channel for regulatory data without

requiring new proprietary systems. HHS should also fund the development of a Healthcare Regulatory Compliance Benchmark dataset — a publicly available, de-identified dataset of compliance scenarios and outcomes that AI compliance tools can use for evaluation and validation, addressing the evaluation gap identified in Q4.

NOETED Response**Patient challenges AI can address.**

NOETED's compliance intelligence focus gives us a specific perspective on patient-facing AI: the compliance failures that result from inadequate regulatory intelligence are not abstract organizational problems — they directly harm patients. When a healthcare organization fails a CMS compliance audit and loses a contract, patients lose access to care providers. When a health plan fails state regulatory requirements and faces sanctions, patients' coverage is disrupted. When an AI compliance tool fails because the federal regulatory data infrastructure it depends on is broken or outdated — as NOETED has documented firsthand — the downstream consequence is that healthcare organizations operate with compliance gaps they cannot see, and patients bear the cost of the resulting enforcement actions. The compliance infrastructure problem is a patient access problem.

From a patient perspective, the healthcare challenges most amenable to AI assistance include:

- Prior authorization delays — patients and caregivers consistently identify prior authorization as a significant barrier to timely care. AI tools that can process prior authorization requests faster and more consistently would have direct, measurable patient benefit.
- Care coordination across fragmented providers — AI tools that can synthesize patient records across multiple providers and identify care gaps would address one of the most common patient complaints about the healthcare system.
- Health literacy and patient education — AI-powered tools that can translate clinical information into plain language appropriate to a patient's health literacy level and preferred language would improve patient engagement and treatment adherence.
- Benefits navigation — patients, particularly those on Medicaid and Medicare, struggle to understand what their coverage includes and how to access it. AI tools that can explain benefits, identify covered services, and navigate prior authorization requirements would reduce a significant administrative burden on patients and caregivers.

Patient concerns about AI.

The patient concerns about AI that NOETED believes HHS should most directly address are:

- Privacy of health information used to train AI — patients want to know whether their health records are being used to train AI systems, who has authorized that use, and how they can opt out. HHS should require clear, plain-language disclosure of AI training data practices as a condition of any federally funded AI deployment.
- Accountability when AI contributes to a wrong decision — patients want to know who is responsible when an AI tool contributes to a misdiagnosis or inappropriate treatment recommendation. The liability ambiguity identified in Q3 — including the novel source data liability question — is not only a problem for healthcare organizations. It is a patient safety and patient rights issue.
- Bias and equity — patients from minority communities and rural areas have legitimate concerns that AI tools trained primarily on data from urban, majority-population health systems will perform less accurately for them. HHS must require demographic performance disaggregation as a standard evaluation requirement.

Q10

What areas of AI research should HHS prioritize to accelerate AI adoption in clinical care? Are there published findings on AI tool impact and costs/benefits in clinical care?

NOETED Response

AI research priorities HHS should fund.

NOETED recommends HHS prioritize the following research areas, in order of near-term impact:

1. Regulatory data infrastructure architecture — HHS should fund foundational research on the design and implementation of standardized regulatory data publication systems for healthcare, including data models, API specifications, and governance frameworks for maintaining regulatory data currency. This research does not exist in the published literature at any meaningful scale and is the prerequisite for every other priority on this list.
2. Regulatory intelligence AI systems — HHS should fund foundational research on the architecture of centralized regulatory intelligence systems for healthcare, including machine learning approaches to regulatory change detection and NLP methods for extracting structured compliance requirements from unstructured regulatory documents.
3. AI evaluation methodology for compliance-sensitive applications — research on how to evaluate AI tools that operate in compliance-sensitive clinical contexts, where the ground truth is a regulatory requirement rather than a clinical outcome, and where source data quality is itself a variable that must be assessed.
4. Demographic performance disaggregation methods — research on statistical methods for identifying and quantifying performance disparities in clinical AI across race, ethnicity, language, geography, and socioeconomic status. Aggregate performance metrics mask disparities that are clinically and legally significant.
5. Human-AI collaboration in clinical workflows — how clinicians actually use AI recommendations in practice, including when and why they override AI recommendations, and what the downstream outcome effects are.
6. AI adoption economics in resource-constrained settings — research specifically addressing the cost-benefit calculus for AI adoption in rural hospitals, FQHCs, and smaller health plans, where the economics differ significantly from large urban health systems.

10a. Published findings on AI tool impact.

The published literature on clinical AI impact shows strong performance in specific, narrow applications — ambient documentation (with studies showing meaningful reductions in physician administrative burden, with some health systems reporting time savings of 25% or more across documentation and administrative tasks), imaging AI for specific diagnostic tasks (radiology and pathology showing sensitivity and specificity competitive with specialists in controlled settings), and sepsis prediction models (with some studies suggesting potential mortality reduction in combination with effective clinical response protocols, though evidence remains mixed). The literature is substantially weaker on real-world deployment outcomes as opposed to controlled evaluation performance, on performance in diverse or underserved populations, and on compliance-adjacent AI applications. The near-complete absence of published research on AI regulatory intelligence systems reflects the infrastructure gap NOETED has identified throughout this comment — you cannot publish research on AI systems that depend on data infrastructure that does not yet exist in reliable form.

10b. Costs, benefits, and transfers in the literature.

The health economics literature on clinical AI is nascent and methodologically inconsistent. Studies vary significantly in how they measure costs (development, deployment, training, ongoing monitoring) and benefits (outcome improvements, administrative efficiency, error reduction). The most consistent finding is that AI tools with narrow, well-defined applications in high-volume workflows show the clearest cost-benefit profiles, while broad-scope AI tools with diffuse benefit claims are harder to evaluate economically.

A critical gap in the literature is the cost of non-adoption: the economic burden of compliance failures, administrative inefficiency, and avoidable clinical errors that result from the absence of reliable AI compliance tools. HHS should fund research that quantifies this counterfactual cost. The scale of the problem is not in dispute: HHS-OIG's own semiannual reports document over \$7 billion annually in expected recoveries from healthcare fraud and abuse investigations, and CMS improper payments in Medicare alone exceeded \$19 billion in FY2024. NOETED's market analysis indicates that a meaningful portion of these failures are attributable to regulatory intelligence gaps — failures caused in part by the broken data infrastructure this comment describes — that better compliance monitoring infrastructure could have prevented. Quantifying that specific causal share would fundamentally change the investment calculus for federal regulatory data infrastructure modernization: it would no longer appear as a discretionary technology investment but as a cost-avoidance measure with a calculable return.

Conclusion

NOETED appreciates the opportunity to contribute to HHS's thinking on AI in clinical care. Our responses across these ten questions return consistently to a single foundational theme: the regulatory data infrastructure problem.

Healthcare AI cannot scale responsibly until healthcare organizations can reliably know what compliance requires of them — in real time, across all applicable federal, state, and local regulatory sources. That knowledge depends on infrastructure: on APIs that work, on datasets that are current, on publication systems that are standardized and maintained. As NOETED has documented through direct experience, that infrastructure does not currently exist in the U.S. healthcare system in any reliable form. The consequences are not abstract — they flow directly to patients who lose access to care when compliance failures result in enforcement actions that could have been avoided.

The sequence of action we recommend is clear: modernize the federal regulatory data publication infrastructure first; establish standardized data formats second; build the certification and safe harbor frameworks that depend on that infrastructure third; and fund the AI research that requires reliable regulatory data to be meaningful fourth. These are not parallel tracks — they are a sequenced infrastructure investment that HHS is uniquely positioned to lead.

NOETED is building the private sector solution to this problem. We are an early-stage company, but we are building at the right moment and in the right direction. We actively seek partnership with ONC, HHS, and aligned academic and non-profit institutions to pilot NOETED's regulatory intelligence platform in real healthcare settings and to contribute to the development of the regulatory data standards the field needs.

We welcome direct engagement with ONC and are available to discuss any of the recommendations in this comment in greater detail.

Respectfully submitted,

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